

Homework 6

Exercises

1. Problem #2 from Section 4.3 of *Mathematical Reasoning: Writing and Proof* [1].
2. Find $\gcd(a, b)$ and express it as a linear combination of a and b .
 - (a) $a = 775, b = 682$.
 - (b) $a = 562, b = 243$.
3. Let $a, b \in \mathbb{Z}$, set $d = \gcd(a, b)$, and consider the set

$$S = \{n \in \mathbb{Z} : n = ax + by \text{ for some } x, y \in \mathbb{Z}\}.$$

Prove that $S = d\mathbb{Z}$, where the right hand side is the usual set of all integer multiples of d .

*Note: This shows that $\gcd(a, b)$ is the **smallest** positive linear combination of a and b .*

4. Let $a, b, c \in \mathbb{Z}^+$ and suppose that $\gcd(a, c) = \gcd(b, c) = 1$. Prove that $\gcd(ab, c) = 1$.

Hint: Problem #3 teaches us that if $m, n, x, y \in \mathbb{Z}$ and $mx + ny = 1$ then $\gcd(m, n) = 1$. So you can solve this problem by using the Bézout identity and then finding a linear combination of ab and c that sums to 1!

5. (Bonus) Explain the error in the following strong induction proof:

Claim: For any $a \in \mathbb{R}^+$, $a^n = 1$ for all $n \in \mathbb{N}$.

Proof. **Base case:** $a^0 = 1$ holds for all $a \in \mathbb{R}^+$.

Inductive step: Fix $n \in \mathbb{N}$ and assume the claim is true, i.e., $a^k = 1$, for all $0 \leq k \leq n$. We want to show that the claim is true for the $n + 1$ case. Observe that

$$a^{n+1} = \frac{a^n \cdot a^n}{a^{n-1}} = \frac{1 \cdot 1}{1} = 1,$$

where we have used the induction hypothesis in the middle equality. Thus the claim is true for the $n + 1$ case and by the Principal of Mathematical Induction we can now conclude that the claim is true for all $n \in \mathbb{N}$. $\square$

6. (Bonus) Explain how the Well-Ordering Principle is related to the Principles of Mathematical Induction.
7. (Bonus) Let $A \subseteq \mathbb{Z}$ be non-empty. Use the Well-Ordering Principle to prove the following Propositions:
 - (a) If A is **bounded below**, meaning there is some $\ell \in \mathbb{Z}$ such that every $a \in A$ satisfies $\ell \leq a$, then there is a **minimum** element $x \in A$ (i.e., x also satisfies $x \leq a$ for all $a \in A$).
Hint: To prove an element $x \in A$ is a minimum of A , you should take $a \in A$ with $a \leq x$ and use principles of deduction to show that we must always have $a = x$.
 - (b) If A is **bounded above**, meaning there is some $u \in \mathbb{Z}$ such that every $a \in A$ satisfies $a \leq u$, then there is a **maximum** element $y \in A$ (i.e., y also satisfies $a \leq y$ for all $a \in A$).

References

- [1] Ted Sundstrom. *Mathematical Reasoning: Writing and Proof*. Grand Valley State University Libraries, 3rd edition, 2020.